PDAUGMENT: DATA AUGMENTATION BY PITCH AND DURATION ADJUSTMENTS FOR AUTOMATIC LYRICS TRANSCRIPTION

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ABSTRACT

Automatic lyrics transcription (ALT), which can be regarded as automatic speech recognition (ASR) on singing voice, is an interesting and practical topic in academia and industry. ALT has not been well developed mainly due to the dearth of the paired singing voice and lyrics datasets for model training. Considering that there is a large amount of ASR training data, a straightforward method is to leverage ASR data to enhance ALT training. However, the improvement is marginal when training the ALT system directly with ASR data, because of the gap between the singing voice and standard speech data which is rooted in music-specific acoustic characteristics in singing voice. In this paper, we propose PDAugment, a data augmentation method that adjusts pitch and duration of speech at syllable level under the guidance of music scores to help ALT training. Specifically, we adjust the pitch and duration of each syllable in natural speech to those of the corresponding note extracted from music scores, to narrow the gap between natural speech and singing voice. Experiments on DSing30 and Dali corpus show that the ALT system equipped with our PDAugment outperforms previous state-of-the-art systems by 5.9% and 18.1% WERs respectively, demonstrating the effectiveness of PDAugment for ALT.

1. INTRODUCTION

Automatic lyrics transcription (ALT), which recognizes lyrics from singing voice, is useful in many applications, such as lyrics-to-music alignment, query-by-singing, karaoke performance evaluation, keyword spotting, and so on. ALT on singing voice can be regarded as a counterpart of automatic speech recognition (ASR) on natural speech. Although ASR has witnessed rapid progress and brought convenience to people in daily life in recent years [1], there is not an ALT system that has the same level of high accuracy and robustness as the current ASR systems. The main challenge of developing a robust ALT system is the scarcity of available paired singing voice and lyrics datasets that can be used for the ALT model training. Though a straightforward method is to use speech data to enhance the training data of ALT, the performance gain is marginal because there are significant differences between speech and singing voice. For example, the singing voice has some music-specific acoustic characteristics [2,3] – the large variation of syllable duration and highly flexible pitch contours are very common in singing, but rarely seen in speech [4].

Previous works have already made some attempts to artificially generate "song-like" data from speech for model training. Kruspe et al. [5] applied time stretching and pitch shifting to natural speech in a random manner, which enriches the distribution of pitch and duration in "songified" speech data to a certain extent. Nonetheless, the adjustments are random, so there is still a gap between the patterns of "songified" speech data and those of real singing voices. Compared to the work of Kruspe et al. [5], Basak et al. [6] further took advantage of real singing voice data. It transferred natural speech to singing voice domain with the guidance of real opera data. But it only took the pitch contours into account, ignoring duration, another key characteristic. Besides, it directly replaced the pitch contours with those of the real opera data, without considering the alignment of the note and syllable, which may result in the low quality of synthesized audio.

In this paper, we propose PDAugment, a syllable-level data augmentation method by adjusting pitch and duration under the guidance of music scores to generate more consistent training data for ALT training. To narrow the gap between the adjusted speech and singing voice, we adjust the speech at the syllable level to make it more in line with the characteristics of the singing voice. We try to make the speech more closely fit the music score, to achieve the effect of "singing out" the speech content. PDAugment adjusts natural speech data by the following steps: 1) extracts note information from music scores to get the pitch and duration patterns of music; 2) aligns the speech and notes at

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syllable level; 3) adjusts the pitch and duration of syllables in natural speech to match those in aligned notes.

Our contributions can be summarized as follows:

- We develop PDAugment, a data augmentation method to enhance the training data for ALT training, by adjusting the pitch and duration of natural speech at syllable level with note information extracted from real music scores.
- We conduct experiments in two singing voice datasets: DSing30 dataset and Dali corpus. The adjusted LibriSpeech corpus is combined with the singing voice corpus for ALT training. The ALT system with PDAugment outperforms the previous state-of-the-art system and Random Aug by 5.9% and 7.8% WERs respectively in DSing30 dataset and 24.9% and 18.1% WERs respectively in Dali corpus. Compared to adding ASR data directly into the training data, our PDAugment has 10.7% and 21.7% WERs reduction in two datasets.
- We analyze the adjusted speech by statistics and visualization and find that PDAugment can significantly compensate the gap between speech and singing voice as well as synthesize relatively high-quality audio.

2. BACKGROUND

2.1 Automatic Lyrics Transcription

The lyrics of a song provide the textual information of singing voice and are important when contributing to the emotional perception of listeners. Automatic lyrics transcription (ALT) aims to recognize lyrics from singing voice, which is of great importance in music information retrieval and analysis. However, ALT is more challenging than ASR – not like speech, in singing voice, the same lyrics with different melodies will have different pitches and durations, which results in the sparsity of training data.

Some works took advantage of the characteristics of music itself: Gupta et al. [7] extended the length of pronounced vowels in output sequences by increasing the probability of a frame with the same phoneme after a certain vowel frame. Kruspe et al. [2] boosted the ALT system by using the newly generated alignment of singing and lyrics. Gupta et al. [8] tried to make use of the background music as extra information to improve recognition accuracy. Ahlback et al. [9] proposed to tag recordings with non-vocal silence and music labels to improve the recognition rate. However, they just designed some hard constraints or added extra information from the music domain, and still did not solve the problem of data scarcity.

Considering the lack of singing voice database, some work aimed at providing a relatively large singing voice dataset: Dabike et al. [10] collected DSing dataset from real-world user information. Demirel et al. [11] built a cascade pipeline with convolutional time-delay neural networks with self-attention based on DSing30 dataset and provided a baseline for ALT task. Some other works leveraged natural speech data for ALT training: they based on pre-trained automatic speech recognition models and then made some adaptations to improve the performance of singing voice: Fujihara et al. [12] built a language model with lyrics but only used the semantic information and ignored the acoustic properties of singing voice. Mesaros et al. [3] used speaker adaptation technique by shifting the GMM components only with global statistics but did not consider the local information.

However, singing voice has some music-specific acoustic characteristics which are not in speech, limiting the performance when training the ALT system directly with natural speech data. Some work tried to synthesize "song-like" data from natural speech to make up for this gap: Kruspe et al. [5] generated "songified" speech data by time stretching, pitch shifting and adding vibrato. However, the degrees of these adjustments are randomly selected within a range, without using the patterns in real music. Basak et al. [6] took use of the F0 contours in real opera data and converted the speech to singing voice through style transfer. Specifically, they decomposed the F0 contours from the real opera data, obtained the spectral envelope and the aperiodic parameter from the natural speech, and then used these parameters to synthesize the singing voice version of the original speech. Nonetheless, in real singing voice, the note and the syllable are often aligned, but they [6] did not perform any alignment of the F0 contours of singing voice with the speech signal. This misalignment may lead to the change of pitch within a consonant phoneme (in normal circumstances, the pitch only changes between two phonemes or in vowels), which further causes distortion of the synthesized audio and limits the performance of ALT system. Besides, they only adjusted the F0 contours, which is not enough to narrow the gap between speech and singing voice.

Some previous works [13, 14] investigated how to convert speech to singing, however, they not only required speech-singing paired data but also a model training process, which made these methods inappropriate for real-time data augmentation.

In this paper, we propose PDAugment, which improves the above adjustment methods by using real music scores and syllable-level alignment to adjust the pitch and duration of natural speech, to solve the problem of insufficient singing voice data.

2.2 Speech vs. Singing Voice

In a sense, singing voice can be considered a special form of speech, but there are still a lot of discrepancies between them [15, 16]. These discrepancies make it inappropriate to transcribe the lyrics from singing voice using a speech recognition model trained on ASR data directly. In order to demonstrate the discrepancies, we randomly select 10K sentences from LibriSpeech [17] for speech corpus and Dali [18] for singing voice dataset, and make some statistics on them. The natural speech and singing voice mainly differ in the following aspects and the analysis results are

listed in Table 1.

2.2.1 Pitch

We extract the pitch contours from singing voice and speech, and compare the range and stability of the pitch. Here we use semitone as the unit of pitch.

Pitch Range Generally speaking, the range of the pitch in singing voice is larger than that in speech. Loscos et al. [15] have pointed out that the frequency range in singing voice can be much larger compared to that in speech. For each sentence, we calculate the pitch range (the maximum pitch value minus the minimum pitch value in this sentence). After averaging the pitch range of the overall 10K sentences in the corpus, the average values are listed in *Pitch Range* in Table 1.

Pitch Stability The pitch of each frame in a certain syllable (when it is corresponding to a note) in singing voice remains almost constant whereas in speech the pitch changes freely along with the audio frames in a syllable. We call the characteristic of maintaining local stability within a syllable as *Pitch Stability*. Specifically, we calculate the pitch difference between every two adjacent frames in a sentence and average it across the entire corpus of 10K sentences. The larger the value of *Pitch Stability*, the more stable the pitch. The results can be seen in *Pitch Stability* of Table 1.

2.2.2 Duration

We also analyze and compare the range and stability of the syllable duration in singing voice and speech. The duration of each syllable varies a lot along with the melody in singing voice. While in speech, it depends on the pronunciation habits of the certain speaker.

Duration Range For each sentence, we calculate the difference between the duration of the longest syllable and the shortest syllable as the duration range. The average values of the duration ranges in the entire corpus are shown as *Duration Range* in Table 1.

Duration Variance We calculate the variance of the duration of syllables in each sentence and average the variances of all sentences in the whole corpus. The results are listed as *Duration Variance* in Table 1 to reflect the flexibility of duration in singing voice.

Property	*Speech*	*Singing Voice*
Pitch Range (semitone)	12.71	14.61
Pitch Stability	0.93	0.84
Duration Range (s)	0.44	2.40
Duration Variance	0.01	0.11

Table 1: The differences of acoustic properties between speech and singing voice.

Besides the differences in the characteristics we mentioned above, sometimes singers may add vibrato in some long vowels or make artistic modifications to the pronunciation of some words to make them sound more melodious, though it will result in a loss of intelligibility. Considering that some characteristics are hard to be quantified, in this work, we start with pitch and duration to build a prototype and propose PDAugment to augment ALT training data by adjusting pitch and duration of speech at syllable level according to music scores.

3. METHOD

3.1 Pipeline Overview

For automatic lyrics transcription, we follow the practice of existing automatic speech recognition systems and choose Conformer encoder [19] and Transformer decoder [20] as our basic model architecture. Different from standard ASR systems, we add a PDAugment module in front of the encoder as shown in Figure 1, to apply syllable-level adjustments to the pitch and duration of the input natural speech according to the information of aligned notes. When the input of ALT system is singing voice, we just do not enable PDAugment module. When the input is speech, PDAugment module takes the note information extracted from music scores as extra input to adjust the pitch and duration of speech, and then adds the adjusted speech into training data to enhance the ALT model. The loss function

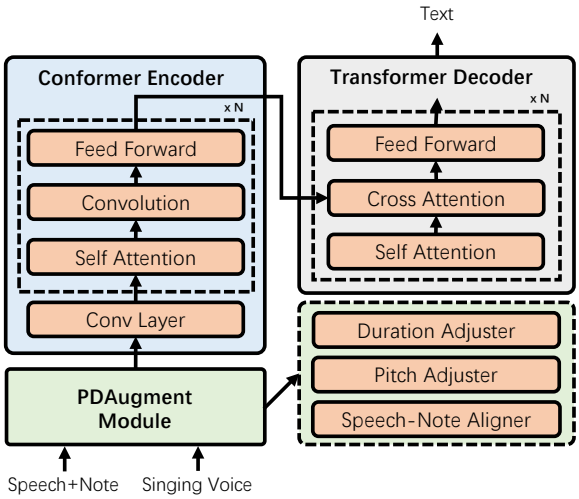


Figure 1: The overall pipeline of the ALT system equipped with PDAugment.

of the ALT model consists of decoder loss $\mathcal{L}_{dec}$, and ctc loss (on top of the encoder) $\mathcal{L}_{ctc}$: $\mathcal{L} = (1-\lambda)\mathcal{L}_{dec}+\lambda\mathcal{L}_{ctc}$, where λ is a hyperparameter to trade off the two loss terms. Considering that lyrics may contain more musical-specific expressions which are rarely seen in natural speech, the probability distributions of lyrics and standard text are quite different. We train the language model with in-domain lyrics data and then fuse it with ALT model in the beam search of the decoding stage.

We try to make the speech fit the patterns of singing voice more naturally as well as to achieve the effect of "singing out" the speech by PDAugment, so we adjust the pitch and duration of speech at syllable level according to those of corresponding notes in music scores instead of applying random adjustments. To do so, we propose PDAugment module, which consists of three key components:

1) speech-note aligner, which generates the syllable-level alignment to decide what the corresponding note of a certain syllable is for subsequent adjusters; 2) pitch adjuster, which adjusts the pitch of each syllable in speech according to that of aligned notes; and 3) duration adjuster, which adjusts the duration of each syllable in speech to be in line with the duration of the corresponding notes. We introduce each part in the following subsections.

3.2 Speech-Note Aligner

According to linguistic and musical knowledge, in singing voice, syllables can be viewed as the smallest textual unit corresponding to notes in the melodies. PDAugment adjusts the pitch and duration of natural speech at syllable level under the guidance of note information obtained from the music scores. In order to apply the syllable-level adjustments, we propose a speech-note aligner, which aims to align the speech and note (in melody) at syllable level.

The textual content can serve as the bridge of aligning notes of melody with the speech. Specifically, our speech-note aligner aligns the speech with notes (in melody) in the following steps:

1) In order to obtain the syllable-level alignment of text and speech, we first convert the text to phoneme using phonemizer [1] and then align the text with speech audio using the Montreal forced aligner (MFA) [21] tool [2] at phoneme level. Next, we group several phonemes into a syllable according to the linguistic rules [22] and get the syllable-level alignment of text and speech.

2) For the syllable-to-note mappings, we set one syllable to correspond to one note by default, because in most cases of singing voice, one syllable is aligned with one note [23]. Only when the time length ratio of the syllable in speech and the note in melody exceeds the predefined thresholds (the lower bound of the ratio is set to 0.5 as the upper bound to 2 from the perspective of preventing severe distortion of the sound), we generate one-to-many or many-to-one mappings to prevent audio distortion after adjustments.

3) We aggregate the syllable-level alignment of text and speech, and the syllable-to-note mappings to generate the syllable-level alignment of speech and note (in melody) as the input of the pitch and duration adjusters.

3.3 Pitch Adjuster

Pitch adjuster adjusts the pitch of input speech at syllable level according to the aligned notes. Specifically, we use WORLD [24], a fast and high-quality vocoder-based speech synthesis system to implement the adjustment. The WORLD system [3] parameterizes speech into three components: fundamental frequency (F0), aperiodicity, and spectral envelope and can reconstruct the speech with only estimated parameters. We use WORLD to estimate the three parameters of natural speech and only adjust the F0 contours according to that of corresponding notes. Then we synthesize speech with adjusted F0 accompanied by original aperiodicity and spectral envelope. Figure 2 shows the F0 contours before and after the pitch adjuster.



Figure 2: The change of F0 contour after pitch adjuster. The content of this example is “opening his door”.

Pitch adjuster calculates the pitch difference between speech and note with syllable-level alignment and adjusts F0 contours of speech accordingly. Some details are as follows: 1) Considering that the quality of synthesized speech will drop sharply when the range of adjustment is too large, we need to keep it within a reasonable threshold. Specifically, we calculate the average pitch of the speech and the corresponding melody respectively. When the average pitch of speech is too different from that of the corresponding melody (e.g., exceeding a threshold, which is 5 semitones in our experiment), we shift the pitch of the entire note sequence to make the difference within the threshold and use the shifted note for adjustment, otherwise, keep the pitch of the original note unchanged; 2) To maintain smooth transitions in synthesized speech and prevent speech from being interrupted, we perform pitch interpolation for the frames between two syllables. 3) When a syllable is mapped to multiple notes, we segment the speech of this syllable in proportion to the duration of notes and adjust the pitch of each segment according to the corresponding note.

3.4 Duration Adjuster

Duration adjuster changes the duration of input speech to align with the duration of the corresponding note. As shown in Figure 3, instead of scaling the whole syllable, we only scale the length of vowels and keep the length of consonants unchanged, because the duration of consonants in singing voice is not significantly longer than that in speech, while long vowels are common in singing voice [5]. There are one-to-many mappings and many-to-one mappings in the syllable-level alignment. For the first case, we calculate the total length of the syllables and adjust the length of all vowels in these syllables in proportion. For the second case, we adjust the length of the vowel, so that the total length of this syllable is equal to the total length of these notes.

[1] https://github.com/bootphon/phonemizer
[2] https://github.com/MontrealCorpusTools/Montreal-Forced-Aligner
[3] https://github.com/JeremyCCHsu/Python-Wrapper-for-World-Vocoder

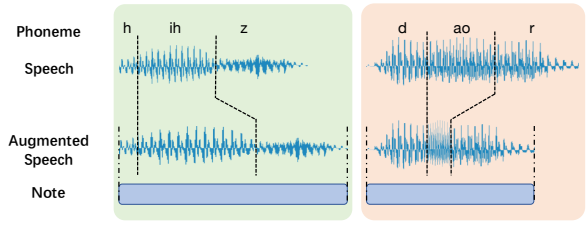


Figure 3: The change of duration after duration adjuster. The content of this example is "his door". The left block shows the case of lengthening the duration of speech and the right shows the case of shortening the duration.

4. EXPERIMENTS AND RESULTS

In this section, we first describe the experimental settings, including datasets, model configuration, and the details of training, inference, and evaluation. Then we report the experiment results, visualize the effect of PDAugment, and further conduct more analyses to verify the effectiveness of PDAugment.

4.1 Experimental Settings

4.1.1 Datasets

Singing Voice Datasets We conduct experiments on two singing voice datasets to verify the effectiveness of our PDAugment: DSing30 dataset [10] and Dali corpus (v2.0) [18, 25]. DSing30 dataset consists of about 4K monophonic Karaoke recordings of English pop songs with nearly 80K utterances, performed by 3,205 singers. We use the partition provided by [10] to make a fair comparison with them. Dali corpus is another large dataset of synchronized audio, lyrics, and notes. It consists of 1200 English polyphonic songs (with background music) for a total duration of 70 hours. Following [6], we use the sentence-level annotation of the dataset provided by [18] [4] and divide the dataset into training, development, and test with a proportion of 8:1:1 without any singers overlapping in each partition. We convert all the singing voice waveforms in our experiments into mel-spectrogram following [26] with a frame size of 25 ms and hop size of 10 ms.

Natural Speech Dataset Following the common practice in previous ASR work [19], we choose the widely used LibriSpeech [17] corpus as the natural speech dataset in our experiments and use the official training partition. The LibriSpeech corpus contains 960 hours of speech sampled at 16 kHz with 1129 female speakers and 1210 male speakers. Similar to singing voice, we convert the speech into mel-spectrogram with the same setting [27].

Music Score Dataset In this work, we choose the pop music subset of FreeMidi dataset [5] because almost all of the songs in our singing voice datasets are pop music. The pop music subset of FreeMidi has about 4000 MIDI files, which are used to provide note information for PDAugment module.

[4] https://github.com/gabolsgabs/DALI
[5] https://freemidi.org/genre-pop

Lyrics Corpus In order to construct a language model with more in-domain knowledge, we deliberately collected a large amount of lyrics data to build our lyrics corpus for language model training. Besides the training text of DSing30 dataset and Dali corpus, we collect lyrics of English pop songs from the Web. We crawl about 46M lines of lyrics and obtain nearly 17M sentences after removing duplicates including DSing30 dataset and Dali corpus. We attach a subset of the collected lyrics corpus in the supplementary material.

4.1.2 Model Configuration

We choose Conformer encoder [19] and Transformer decoder [20] as the basic ALT model architecture in our experiments since the effectiveness of the structure has been proved in ASR. Our language model is based on Transformer encoder. More details about the model configuration, training/inference settings, and codes are attached in the supplementary materials.

4.2 Results

In this subsection, we report the experimental results of the ALT system equipped with PDAugment in two singing voice datasets. We compare our results with several basic settings as baselines: 1) *Naive ALT*, the ALT model trained with only singing voice dataset; 2) *ASR Augmented*, the ALT model trained with the combination of singing voice dataset and ASR data directly; 3) *Random Aug* [5], the ALT model trained with the combination of singing voice dataset and randomly adjusted ASR data. The pitch is adjusted ranging from -6 to 6 semitones randomly and the duration ratio of speech before and after adjustment is randomly selected from 0.5 to 1.2. All 1), 2), and 3) are using the same model architecture as *PDAugment*. Besides the above three baselines, we compare our PDAugment with the previous systems which reported the best results in two datasets respectively. We compare our PDAugment with [11] using RNNLM for DSing30 dataset and compare the results with [6] for Dali corpus, All the results are listed in Table 2.

Method	DSing30		Dali	
	Dev	*Test*	*Dev*	*Test*
Naive ALT	28.2	27.4	80.9	86.3
ASR Augmented	20.8	20.5	75.5	75.7
Random Aug [5]	17.9	17.6	69.8	72.1
Previous Work [11]	17.7	15.7	-	-
Previous Work [6]	-	-	75.2	78.9
PDAugment	**10.1**	**9.8**	**53.4**	**54.0**

Table 2: The WERs (%) of DSing30 and Dali dataset. The audios of Dali corpus contain background music.

As can be seen, *Naive ALT* performs not well and gets high WERs in both DSing30 dataset and Dali corpus, which demonstrates the difficulty of the ALT task. After adding ASR data for ALT training, the performances of *ASR Augmented* setting in both of the two datasets have

been improved slightly compared to *Naive ALT*, but still with relatively high WERs, which indicates the limitation of using ASR training data directly.

When applying the adjustments, a question is if we adjust the pitch and duration with random ranges without note information from music scores, how well will the ALT system perform? The results of *Random Aug* can perfectly answer this question. As the results show, *Random Aug* can slightly improve the performance compared with *ASR Augmented*, demonstrating that increasing the volatility of pitch and duration in natural speech helps ALT system training, which is the same as what [5] claimed. *PDAugment* is significantly better than *Random Aug*, which indicates that adjusted speech can better help the ALT training with the guidance of music scores.

Besides, it is obvious that *PDAugment* greatly outperforms the previous work in both datasets. [11] performs worse than *PDAugment* because of not taking advantage of the massive ASR training data. Compared with [6] that replaced F0 contours of speech directly, *PDAugment* can narrow the gap between natural speech and singing voice in a more reasonable manner and achieve the lowest WERs among all the above methods. The results in both datasets show the effectiveness of PDAugment for ALT task and reflect the superiority of adding music-specific acoustic characteristics into natural speech.

4.3 Ablation Studies

We conduct more experimental analyses to deeply explore our PDAugment and verify the necessity of some design details. More ablation studies (different language models) can be found in the supplementary materials. The ablation studies are carried out on DSing30 dataset in this section.

4.3.1 Augmentation Types

In this subsection, we explore the effects of different kinds of augmentations (only adjust pitch, only adjust duration, and adjust pitch & duration) on increasing the performance of ALT system. We generate three types of adjusted speech by enabling different adjusters of PDAugment module and conduct the experiments on DSing30. The results are shown in Table 3.

Setting	*DSing30 Dev*	*DSing30 Test*
PDAugment	**10.1**	**9.8**
Disable Pitch Adjuster	13.6	13.4
Disable Duration Adjuster	13.8	13.8
Disable both Adjusters	20.8	20.5

Table 3: The WERs (%) of different types of augmentation of DSing30 dataset. All of the settings are trained on DSing30 and the original or adjusted LibriSpeech data.

As we can see, when we enable the whole PDAugment module, the ALT system can achieve the best performance, indicating the effectiveness of the pitch and duration adjusters. When we disable the pitch adjuster, the WER on DSing30 is 3.6% higher than *PDAugment*. The same thing happens when we disable the duration adjuster, the WER is 4.0% higher than *PDAugment*. And if both the pitch and duration are not adjusted, which means using the speech data directly for ALT training, the WER is the worst among all settings. The results demonstrate that both pitch and duration adjusters are necessary and can help with improving the recognition accuracy of the ALT system.

4.4 Adjusted Speech Analyses

Following Section 2.2, we analyze the acoustic properties of the original natural speech and the adjusted speech by random and PDAugment, and list the results in Table 4.

Property	*Original*	*Random*	*PDAugment*
Pitch Range (semitone)	12.71	13.87	14.19
Pitch Stability	0.93	0.59	0.69
Duration Range (s)	0.44	0.42	0.59
Duration Variance	0.01	0.01	0.05

Table 4: The differences of acoustic properties between original speech and adjusted speech.

Combining the information of Table 1 and Table 4, we can clearly find that the distribution pattern of acoustic properties (pitch and duration) after PDAugment is closer to singing voice compared with the original speech, which indicates that our PDAugment can change the patterns of pitch and duration in original speech and effectively narrow the gap between natural speech and singing voice. To avoid the distortion of adjusted speech, we limit the adjustment degree within a reasonable range, so the statistics of adjusted speech can not completely match that of singing voice. Nonetheless, adjusted speech is still good enough for ALT model to capture some music-specific characteristics.

In order to visually demonstrate the effect of PDAugment module, we plot the spectrograms of speech to compare the acoustic characteristics before and after different types of adjustments. The visualization of an example adjusted speech is attached in the supplementary materials.

5. CONCLUSION

In this paper, we proposed PDAugment, a data augmentation method by adjusting pitch and duration to make better use of natural speech for ALT training. PDAugment transfers natural speech into singing voice domain by adjusting pitch and duration at syllable level under the instruction of music scores. PDAugment module consists of speech-note aligner to align the speech with notes, and two adjusters to adjust pitch and duration respectively. Experiments on two singing voice datasets show that PDAugment can significantly reduce the WERs of ALT task. We also explore different types of augmentation, and further verify the effectiveness of PDAugment. In the future, we will consider narrowing the gap between natural speech and singing voice from more aspects such as vibrato, and try to add some music-specific constraints in the decoding stage.

6. ACKNOWLEDGEMENT

This work was supported by the Key Project of Natural Science Foundation of Zhejiang Province (No.LZ19F020002), the Key R&D Program of Zhejiang Province (No.2022C03126), and Project of Key Laboratory of Intelligent Processing Technology for Digital Music (Zhejiang Conservatory of Music), Ministry of Culture and Tourism (No.2022DMKLB001). Thanks to Future Design Laboratory of Zhejiang University for the help in dataset collection. Thanks are also due to Shen Zhou for bringing insights in music.

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